

Monthly Online Test-1 | JEE - 2022

Date: 22/05/2020

Maximum Marks: 300

General Instructions

1. The test is of **3 hours** duration and the maximum marks are **300**.
2. The question paper consists of **3 Parts** (Part I: **Mathematics**, Part II: **Physics**, Part III: **Chemistry**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **Five Integer value type questions**. Each question has an integer value ranging from **0 – 999**.

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Syllabus:

Physics : Vectors**Chemistry** : Atomic Structure**Mathematics** : Quadratic Equation

SUBJECT I : PHYSICS**Suggested Time: 60 Minutes****100 MARKS**

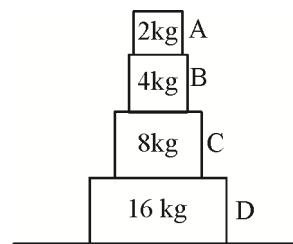
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- A force $(3\hat{i} + 4\hat{j})$ newton acts on a body and displaces it by $(3\hat{i} + 4\hat{j})$ metre. The work done by the force is:
(A) 5J (B) 25 J (C) 10 J (D) 30 J
- If a unit vector is represented by $0.5\hat{i} + 0.8\hat{j} + c\hat{k}$ the value of c is:
(A) 1 (B) $\sqrt{0.11}$ (C) $\sqrt{0.01}$ (D) 0.39
- If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{i} - 4\hat{j} + \alpha\hat{k}$, then the value of α is:
(A) -1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) 1
- The angle between the two vector $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ is :
(A) 60° (B) 45° (C) 90° (D) 30°
- Unit vector parallel to the resultant of vectors $\vec{A} = 4\hat{i} - 3\hat{j}$ and $\vec{B} = 8\hat{i} + 8\hat{j}$ will be:
(A) $\frac{24\hat{i} + 5\hat{j}}{13}$ (B) $\frac{12\hat{i} + 5\hat{j}}{13}$ (C) $\frac{6\hat{i} + 5\hat{j}}{13}$ (D) None of these
- The component of vector $\vec{A} = 2\hat{i} + 3\hat{j}$ along the vector $\hat{i} + \hat{j}$ is:
(A) $\frac{5}{\sqrt{2}}$ (B) $10\sqrt{2}$ (C) $5\sqrt{2}$ (D) 5
- If $|\vec{A} \times \vec{B}| = \sqrt{3} \vec{A} \cdot \vec{B}$, then the value of $|\vec{A} + \vec{B}|$ is :
(A) $(A^2 + B^2 + AB)^{1/2}$ (B) $(A^2 + B^2 + \frac{AB}{\sqrt{3}})^{1/2}$
(C) $(A + B)$ (D) $(A^2 + B^2 + \sqrt{3}A \cdot B)^{1/3}$
- The value of n so that vectors $2\hat{i} + 3\hat{j} - 2\hat{k}, 5\hat{i} + n\hat{j} + \hat{k}$ and $-\hat{i} + 2\hat{j} + 3\hat{k}$ may be coplanar, will be:
(A) 18 (B) 28 (C) 9 (D) 36
- Three vectors satisfy the relation $\vec{A} \cdot \vec{B} = 0$ and $\vec{A} \cdot \vec{C} = 0$, then $\vec{A}$ is parallel to :
(A) $\vec{C}$ (B) $\vec{B}$ (C) $\vec{B} \times \vec{C}$ (D) $\vec{B} \cdot \vec{C}$
- The angle which the vector $\vec{A} = 2\hat{i} + 3\hat{j}$ makes with the y-axis, where $\hat{i}$ and $\hat{j}$ are unit vectors along x- and y-axis, respectively, is:
(A) $\cos^{-1}(3/5)$ (B) $\cos^{-1}(2/3)$ (C) $\tan^{-1}(2/3)$ (D) $\sin^{-1}(2/3)$
- The angle between $\vec{A} + \vec{B}$ and $\vec{A} \times \vec{B}$ is:
(A) 0 (B) $\pi/4$ (C) $\pi/2$ (D) π

12. What is the angle between $\vec{A}$ and $\vec{B}$, if $\vec{A}$ and $\vec{B}$ are the adjacent sides of a parallelogram drawn from a common point and the area of the parallelogram is $AB/2$?
 (A) 15° (B) 30° (C) 45° (D) 60°

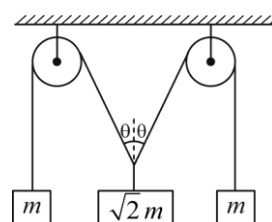
13. Find the normal reaction applied by block C on block B for the given system lying on a stationary horizontal surface. $\{g = 10\text{m/s}^2\}$

- (A) 20 N
 (B) 40 N
 (C) 80 N
 (D) 60 N



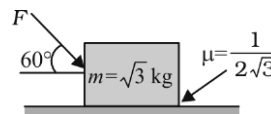
14. The pulley and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be:

- (A) 0° (B) 30°
 (C) 45° (D) 60°

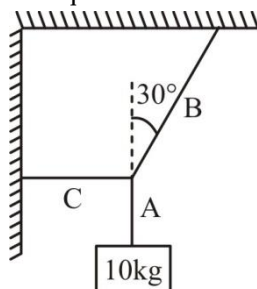


15. What is the maximum value of the force F such that the block shown in the figure does not move?

- (A) 20 N (B) 10 N
 (C) 12 N (D) 15 N



16. In the figure a block of mass 10 kg is in equilibrium. Identify the string in which the tension is zero.

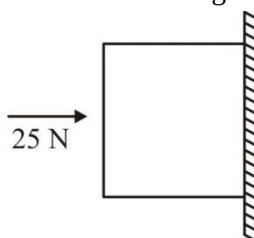


- (A) B (B) C
 (C) A (D) None of the above

17. A block of mass m is placed at rest on an inclined plane of inclination θ to the horizontal. If the coefficient of friction between the block and the plane is μ , then the total force the inclined plane exerts on the block is:

- (A) mg (B) $\mu mg \cos \theta$ (C) $mg \sin \theta$ (D) $\mu mg \tan \theta$

18. A horizontal force of 25 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.4 . The weight of the block is:



- (A) 2.5 N (B) 20 N (C) 10 N (D) 5 N

19. If $\vec{A}$ is perpendicular to $\vec{B}$, then :
- (A) $\vec{A} \times \vec{B} = 0$ (B) $\vec{A} \cdot [\vec{A} + \vec{B}] = A^2$
- (C) $\vec{A} \cdot \vec{B} = AB$ (D) $\vec{A} \cdot [\vec{A} + \vec{B}] = A^2 + AB$
20. In going from one city to another, a car travel 75 km north, 60 km north-west and 20 km east. The magnitude of displacement between the two cities is (take $1/\sqrt{2} = 0.7$)
- (A) 170 km (B) 137 km (C) 119 km (D) 140 km

SECTION 2

This section contains 5 Integer value type questions. Each question has an integer value ranging from 0 – 999.

21. The sum of the magnitudes of two forces acting at a point is 16N. The resultant of these forces is perpendicular to the smaller force and has a magnitude of 8N. If the smaller force is of magnitude x , then the value of x is (in Newton) :
22. If $|\vec{A} + \vec{B}| = |\vec{A}| = |\vec{B}|$, then the angle between $\vec{A}$ and $\vec{B}$ is (in degree):
23. Two vectors $\vec{a}$ and $\vec{b}$ are such that $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$. What is the angle between $\vec{a}$ and $\vec{b}$ (in degree)?
24. Given that $\vec{A} + \vec{B} = \vec{C}$ if $|\vec{A}| = 4$, $|\vec{B}| = 5$ and $|\vec{C}| = \sqrt{61}$, the angle between $\vec{A}$ and $\vec{B}$ is (in degree):
25. Maximum and minimum values of the resultant of two forces acting at a point are 7 N and 3 N respectively. The smaller force will be equal to (in Newton):

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Which of the following statements does not describe Rutherford's atomic model ?
 (A) Atom is mostly made of empty space
 (B) The centre of the atom has positively charged nucleus
 (C) The size of the nucleus is very small as compared to the size of the atom
 (D) The electrons revolve around the nucleus only in certain orbits of fixed energy and radius.
- To which of the following is the energy of electromagnetic radiation inversely proportional?
 I. Frequency II. Wavelength III. Wave number
 (A) Only I and III (B) Only II
 (C) Only II and III (D) None of the above
- Which of the following types of waves has the shortest wavelength?
 (A) Radio waves (B) UV rays (C) X-rays (D) Microwaves
- The energy of green light whose frequency is 6×10^{14} Hz is: ($h = 6.626 \times 10^{-34}$ Js)
 (A) 39×10^{-19} J (B) 3.9×10^{-19} J (C) 1.104×10^{-48} J (D) 11.04×10^{-48} J
- The threshold wavelength for photoelectric effect on sodium is 500 nm. Its work function is: ($C = 3 \times 10^8$ m/s)
 (A) 4×10^{-19} J (B) 1 J (C) 2×10^{-19} J (D) 3×10^{-19} J
- Which of the following is correct? (r_n is radius of n th Bohr's Orbit)
 (A) $r_n = \frac{n^2 h}{4\pi^2 K Z m e^2}$ (B) $r_n = \frac{n^2 h^2}{4\pi^2 K Z m e^2}$
 (C) $r_n = \frac{nh}{4\pi^2 K Z m e^2}$ (D) $r_n = \frac{nh^2}{4\pi K Z m e^2}$
- If the velocity of an electron in the first Bohr orbit in H-atom is v . What will be the velocity of the electron in the 2nd orbit of He^+ ? (Atomic No. of He = 2)
 (A) v (B) $2v$ (C) $v/2$ (D) $4v^2$
- What is the ionization energy of an electron present in the 4th excited state of a H-atom?
 (A) -0.544 eV (B) $+0.85$ eV (C) $+0.544$ eV (D) $+1.51$ eV
- The angular momentum of an electron in H-atom (according to Bohr's theory) present in the first excited state is: (Given $h = 6.626 \times 10^{-34}$ Js, $\pi = 3.14$)
 (A) 2.11×10^{-34} Jsec (B) 3×10^{-34} Jsec
 (C) 20×10^{-34} Jsec (D) 0.5×10^{-34} Jsec
- Hydrogen atoms are excited to $n = 4$ energy state. In the spectrum of emitted radiation, ratio of number of lines in the U.V. and visible regions are respectively:
 (A) 3 : 1 (B) 1 : 3 (C) 2 : 3 (D) 3 : 2

11. Which of the following is not a drawback of the Bohr's Model?
 (A) His postulates combined two different concepts : one from classical physics and second from modern theory of quantization.
 (B) It could explain the spectra of atoms or ions having two or more electrons but not of monoelectronic species.
 (C) It could not provide a satisfactory picture of the chemical bond.
 (D) None of the above
12. If the uncertainty in position and momentum are equal, the uncertainty in velocity is :
 (A) $\sqrt{h/\pi}$ (B) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$
 (C) $\sqrt{h/2\pi}$ (D) Cannot be determined
13. Two particles of masses 'm' and '2m' have equal K.E. Their de-Broglie's wavelengths are in the ratio of:
 (A) 1:1 (B) 1:2 (C) $1:\sqrt{2}$ (D) $\sqrt{2}:1$
14. Which of the following set of quantum numbers is not possible for sodium atom : (Atomic number = 11)

n	ℓ	m_ℓ	m_s		n	ℓ	m_ℓ	m_s
(A) 2	1	-2	$-\frac{1}{2}$		(B) 2	1	-1	$+\frac{1}{2}$
(C) 3	0	0	$+\frac{1}{2}$		(D)	None of these		
15. The respective values of orbital angular momentum of an electron in 1s, 2s, 3d and 3p orbital are :
 (A) $0, 0, \sqrt{6}\frac{h}{2\pi}, \sqrt{2}\frac{h}{2\pi}$ (B) $1, 1, \sqrt{4}\frac{h}{2\pi}, \sqrt{2}\frac{h}{2\pi}$
 (C) $0, 1, \sqrt{6}\frac{h}{2\pi}, \sqrt{3}\frac{h}{2\pi}$ (D) $0, 0, \sqrt{20}\frac{h}{2\pi}, \sqrt{6}\frac{h}{2\pi}$
16. Which of the following is the correct order of energy of orbitals for H-atom ?
 I. $1s < 2p$ II. $2s = 2p$ III. $3d < 4s$
 (A) Only I (B) Only II and III (C) I, II and III (D) None of the above
17. Which of the following is/are correctly matched ?
- | | Orbital | Number of radial nodes | Number of angular nodes |
|-----|---------|------------------------|-------------------------|
| I | 2s | 1 | 0 |
| II | 2p | 1 | 1 |
| III | 3p | 1 | 2 |
- (A) Only II and III (B) Only I
 (C) Only II (D) I, II and III
18. Which of the following is the correct outer most electronic configuration of V^{2+} ? (Atomic number of V = 23)
 (A) $[Ar]3d^3 4s^2$ (B) $[Ar]3d^1 4s^2$ (C) $[Ar]3d^3 4s^0$ (D) $[Ar]3d^5 4s^0$

19. Find the number of waves made by electron in one complete revolution in third Bohr's orbit.
(A) six (B) Four (C) Three (D) Two
20. The maximum number of 3d electrons having spin quantum number $s = +\frac{1}{2}$ are :
(A) 10 (B) 14
(C) 5 (D) any number from 1 to 10

SECTION 2

This section contains 5 Integer value type questions. Each question has an integer value ranging from 0 – 999.

21. What will be the total number of electrons having $n + \ell = 3$ in Cr ($Z = 24$) in its ground state?
22. Number of spectral lines in the visible region lines when an electron returns from 5th orbit to ground state in H spectrum is x . Find the value of x .
23. The wave number of electromagnetic radiation emitted during the transition of electron in between two levels of Li^{2+} ion whose principal quantum number sum is four and difference is 2, is $n \times R_H$. What is the value of n ?
24. What is the degeneracy of fourth principal energy level of H atom?
25. Which Bohr's orbit of Be^{3+} has the same orbit radius as that of the ground state of hydrogen atom? (Atomic number of Be = 4)

SUBJECT III: MATHEMATICS

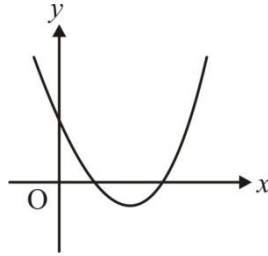
Suggested Time: 75 Minutes

120 MARKS

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then $\frac{1}{a\alpha + b} + \frac{1}{a\beta + b}$ is equal to:
- (A) $\frac{b}{ac}$ (B) $\frac{-b}{ac}$ (C) $\frac{ac}{b}$ (D) $\frac{-ac}{b}$

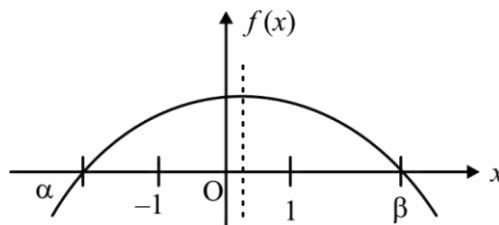
2. The graph of $y = ax^2 + bx + c$ is as shown. Which one of the following is NOT correct?



- (A) $a > 0$ (B) $b^2 - 4ac > 0$ (C) $b < 0$ (D) $abc > 0$
3. Let $P(x) = x^2 - (2 - P)x + P - 2$. If $P(x)$ assumes both positive and negative values for $x \in R$, then the range of P is:
- (A) $(-\infty, 2) \cup (6, \infty)$ (B) $(2, 6)$
(C) $(-\infty, 2)$ (D) $(6, \infty)$
4. If α, β, γ are roots of equation $111x^3 - 11x - 1 = 0$ then $(\alpha\beta)^{-2} + (\beta\gamma)^{-2} + (\gamma\alpha)^{-2}$ equals
- (A) 2332 (B) 1331 (C) 1210 (D) 2442
5. The zeroes of the quadratic polynomial $f(x) = 2x^2 - 3x + k^2 - 3k + 4$, $k \in R$ lie on both sides of the origin then the set of the values of k is:
- (A) ϕ (B) $(-4, 1)$ (C) $(-1, 4)$ (D) $(-\infty, -4) \cup (1, \infty)$
6. If α, β are the roots of $x^2 - p(x+1) - c = 0, c \neq 1$, then the value of $(\alpha+1)(\beta+1)$ is equal to:
- (A) $1 - c$ (B) $p + c$ (C) $1 + c$ (D) $1 - 2p$
7. If α, β are roots of the equation $x^2 + ax + 2 = 0$ such that $\alpha^6\beta^5 = 32$, then the value of 'a' is:
- (A) -2 (B) 2 (C) -3 (D) 3
8. Let $f(x) = ax^2 + bx + c$ ($x \in R, a > 0$) such that $f(2)$ is minimum value of $f(x)$, then :
- (A) $f(2) < f(1) < f(3)$ (B) $f(2) < f(0) < f(5)$
(C) $f(-1) < f(0) < f(1)$ (D) $f(2) < f(0) = f(3)$

9. If roots of the equation $lx^2 + mx + n = 0$ are (2α) and $(\alpha + \beta)$, roots of the equation $px^2 + qx + r = 0$ are (2β) and $(\alpha + \beta)$ and D_1 and D_2 are the respective discriminants, then $\frac{D_1}{D_2}$ is equal to (where $\alpha \neq \beta$) :
- (A) $\frac{lmn}{pqr}$ (B) $\left(\frac{l}{p}\right)^2$ (C) $\left(\frac{m}{q}\right)^2$ (D) $\left(\frac{n}{r}\right)^2$
10. For what value of 'm' roots of equation $x^2 - x + m = 0$ are not real-
- (A) $\left(\frac{1}{4}, \infty\right)$ (B) $\left(-\infty, \frac{1}{4}\right)$ (C) $\left(-\frac{1}{4}, \frac{1}{4}\right)$ (D) None of these
11. The expression $y = ax^2 + bx + c$ has always the same sign as c if-
- (A) $4ac < b^2$ (B) $4ac > b^2$ (C) $ac < b^2$ (D) $ac > b^2$
12. If $c^2 = 4d$ and the two equations $x^2 - ax + b = 0$ and $x^2 - cx + d = 0$ have one common root, then the value of $2(b + d)$ is equal to:
- (A) $\frac{a}{c}$ (B) ac (C) $2ac$ (D) $a + c$
13. If the vertex of parabola $y = 2x^2 - 4x + 6$ is (m, n) , then the parabola whose vertex is $\left(2m, \frac{n}{2}\right)$ is:
- (A) $y = x^2 - 4x + 8$ (B) $y = x^2 + 4x + 6$
 (C) $y = -2x^2 + 8x - 6$ (D) $y = 2x^2 + 8x - 8$
14. If m and n are real and different satisfying the relation $m^2 = 4m + 4$ and $n^2 = 4n + 4$, then $(m^2 + n^2 + 4mn)$ is equal to:
- (A) 2 (B) 4 (C) 8 (D) 16
15. The value of 'a' for which one root of the equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2$ is twice as large as the other is-
- (A) $-\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $-\frac{2}{3}$ (D) $\frac{1}{3}$
16. Two student attempted to solve a quadratic equation $x^2 + bx + c = 0$. Although both student did work correctly. One miscopied the middle term and obtained the solution set $\{2, 3\}$. While the other miscopied the constant term and obtained solution set $\{2, 5\}$ the correct solution set of quadratic equation is-
- (A) $\{1, 6\}$ (B) $\{2, 5\}$ (C) $\{3, 6\}$ (D) $\{1, 4\}$
17. If α, β are roots of $x^2 - 2x + 3 = 0$ then equation whose roots are $\frac{\alpha-1}{\alpha+1}, \frac{\beta-1}{\beta+1}$ will be:
- (A) $3x^2 - 2x + 1 = 0$ (B) $3x^2 + 2x + 1 = 0$ (C) $3x^2 - 2x - 1 = 0$ (D) $x^2 - 3x + 1 = 0$

18. The number of integral value of x satisfying the inequality $5x-1 < (x+1)^2 < 7x-3$, is/are :
 (A) 0 (B) 1 (C) 2 (D) 3
19. If $x^2 - 11x + a = 0$, $x^2 - 14x + 2a = 0$ have one common root then sum of possible value of a is :
 (A) -24 (B) 1 (C) 24 (D) 25
20. The graph of quadratic polynomial $f(x) = ax^2 + bx + c$ is shown below ($|\beta| > |\alpha|$)



which of the following is **INCORRECT**?

- (A) $\frac{c}{a} < -1$ (B) $|\beta - \alpha| > 2$
 (C) $f(x) > 0 \quad \forall x \in (0, \beta)$ (D) $abc > 0$

SECTION 2

This section contains 5 Integer value type questions. Each question has an integer value ranging from 0 – 999.

21. If $x^2 + bx + b$ is a factor of $x^3 + 2x^2 + 2x + c$ ($c \neq 0$), then the value $b - c$ is equal to
22. If α, β are roots of the equation $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + 8 + 2\sqrt{5} = 0$ then $\frac{2}{\frac{1}{\alpha} + \frac{1}{\beta}}$ is equal to-
23. α and β are roots of the equation $2x^2 - 35x + 2 = 0$. Find the value of $\sqrt{(2\alpha - 35)^3 (2\beta - 35)^3}$.
24. The number of all values of n , (where n is a whole number) for which the equation $\frac{x-8}{n-10} = \frac{n}{x}$ has no real solution for x .
25. For real x , then integer which does not lie in the range of $f(x)$, where $f(x) = \frac{x^2 - 2x + 2}{2x - 2}$.